



GramKavach: CyberShield

AI-Powered Pre-PIN Fraud Prevention for Rural India

"Think Before You Tap. AI Already Did."

Presented by

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Problem Statement & AI Solution

THE PROBLEM. THE INNOVATION. THE SOLUTION.

The Digital Reality of Rural India

400+ Million

UPI Users

Millions of first-time users are entering India's digital payment ecosystem.

₹1,087 Crore

Reported Fraud Losses

Based on official Government of India data (FY24).

70%

Rural Users at Risk

A significant share of fraud victims comes from rural and Tier-2/3 regions.

 **Language Barrier:** Many cybersecurity alerts and awareness resources are primarily available in English, making them difficult for many rural users to understand.

Sources: [Ministry of Finance - Parliament Q&A \(March 2026\)](#) • [NPCI Official Data](#) • [RBI Digital Payment Reports](#)

The Silent Digital Crisis

Rising UPI Adoption, Low Literacy

Rapid digital payment adoption, combined with limited digital literacy, creates a significant security gap in rural India.

Language Barrier in Cybersecurity

Many security alerts and awareness resources are primarily available in English, reducing accessibility for many rural users.

Common Fraud Vectors

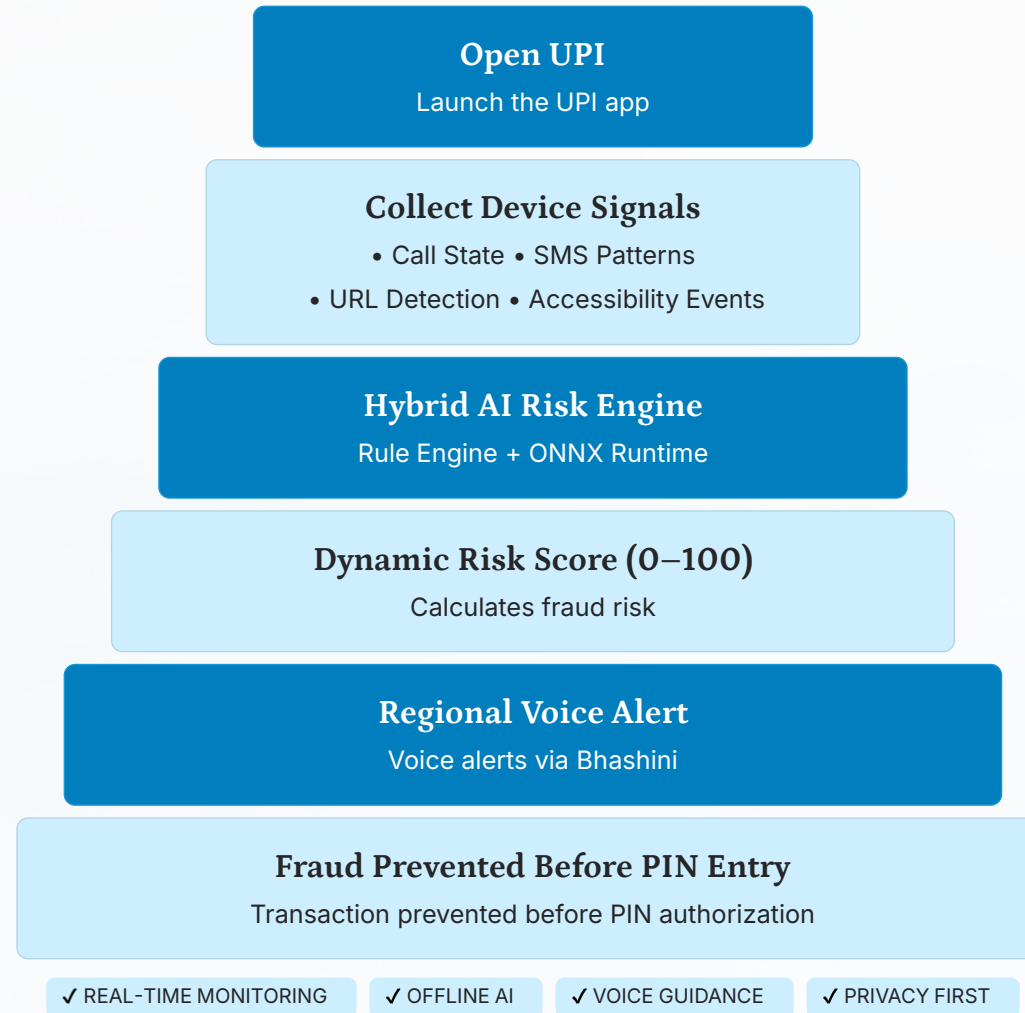
Fake SMS, phishing links, scam calls, remote-access apps, fake QR codes, and fraudulent UPI collect requests are common attack vectors.

Existing Solutions Are Often Reactive

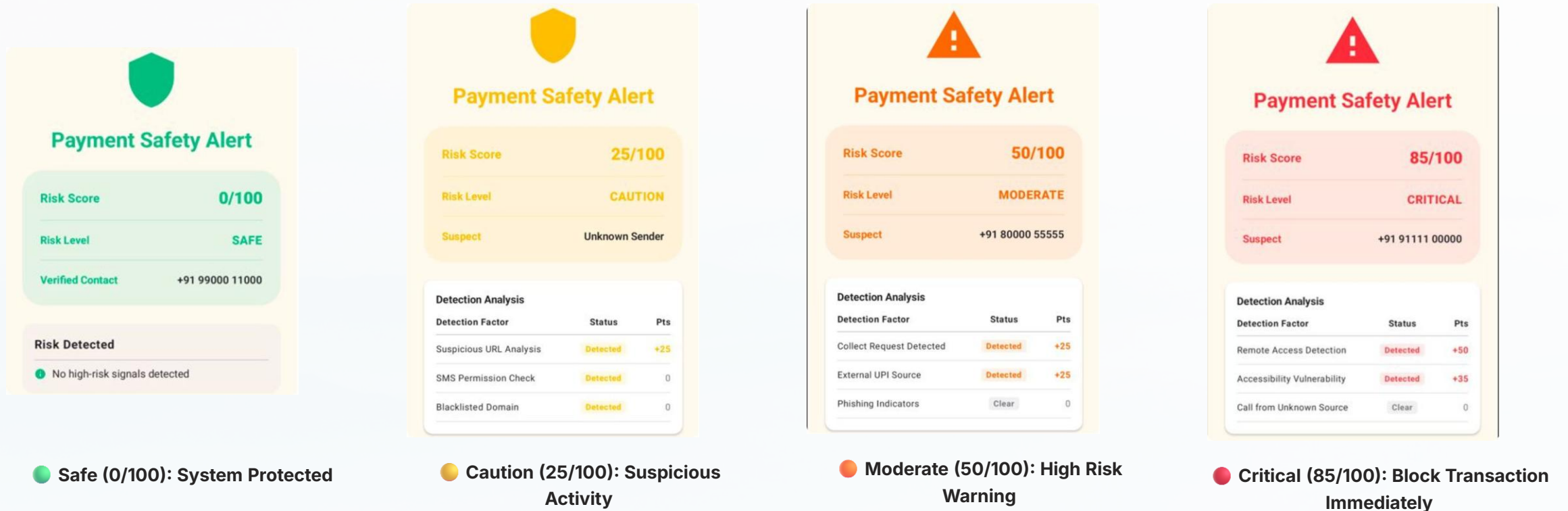
Many solutions focus on detection or recovery after suspicious activity rather than preventing fraudulent transactions before authorization.

Sources [Ministry of Finance - Parliament Q&A \(March 2026\)](#) • [RBI Digital Payment Fraud Report](#) • [Gujarat CID Cyber Cell - Operation Mule Hunt Report \(June 2026\)](#) • [IEEE SIGHT Gujarat - Rural Digital Literacy Study](#)

GramKavach: AI-Powered Pre-PIN Protection



Live AI Risk Assessment



Demo Scenario: As suspicious signals increase, the AI raises the risk score and alerts the user before transaction authorization.

What Makes GramKavach Unique

Conventional Protection

Detects fraud after transaction

Alerts arrive too late, after the money has already moved.

Alerts after suspicious activity

Alerts only after suspicious activity begins.

Financial loss has already occurred

Money already transferred

GramKavach

Detects before PIN entry

Analyzes risk before the user enters the UPI PIN.

Proactive prevention

Provides real-time voice warnings before payment authorization.

Stops fraud at source

Blocks fraudulent transactions before authorization.

Core Strengths

Offline AI

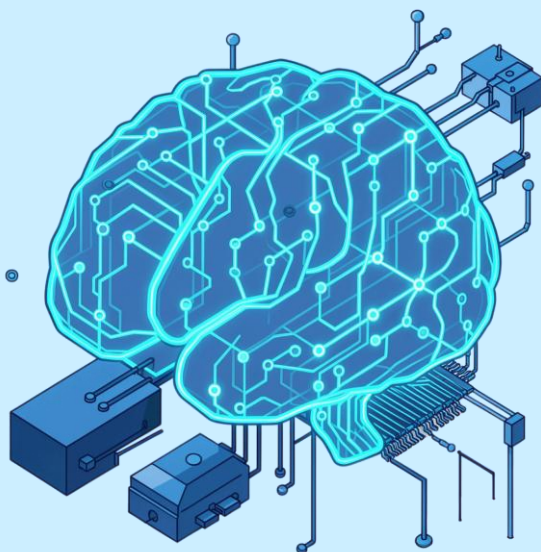
- Runs on-device
- Works without internet
- Privacy-first

Voice Guidance

- Powered by Bhashini APIs
- Multilingual alerts
- Accessible for all users

Technology Stack

Offline-first AI architecture enabling real-time fraud detection and multilingual voice alerts.



Built with

- Kotlin
- Jetpack Compose
- ONNX Runtime
- Bhashini APIs
- Room
- Hilt

Key Features

- Real-time Monitoring
- Offline AI Inference
- Multilingual Voice Alerts
- Privacy First Processing

 100% On-Device • Offline-First • Privacy by Design

Robust & Scalable System Architecture

Clean Architecture

- Multi-module design split into `:domain`, `:data`, `:ai`, `:monitoring`, and `:app`.
Modular architecture with separate Domain, Data, AI, Monitoring, and UI layers for scalability and maintainability.

Zero-Trust Signal Bus

- Unified background monitoring for calls, SMS, and app states in one secure pipeline.
Collects risk signals from calls, SMS, URLs, and app activity for real-time threat detection.

Hybrid AI Engine (ONNX Mobile)

- Microsoft ONNX Runtime executes the AI model directly on-device.
Combines rule-based detection with ONNX Runtime for fast, offline AI risk assessment.

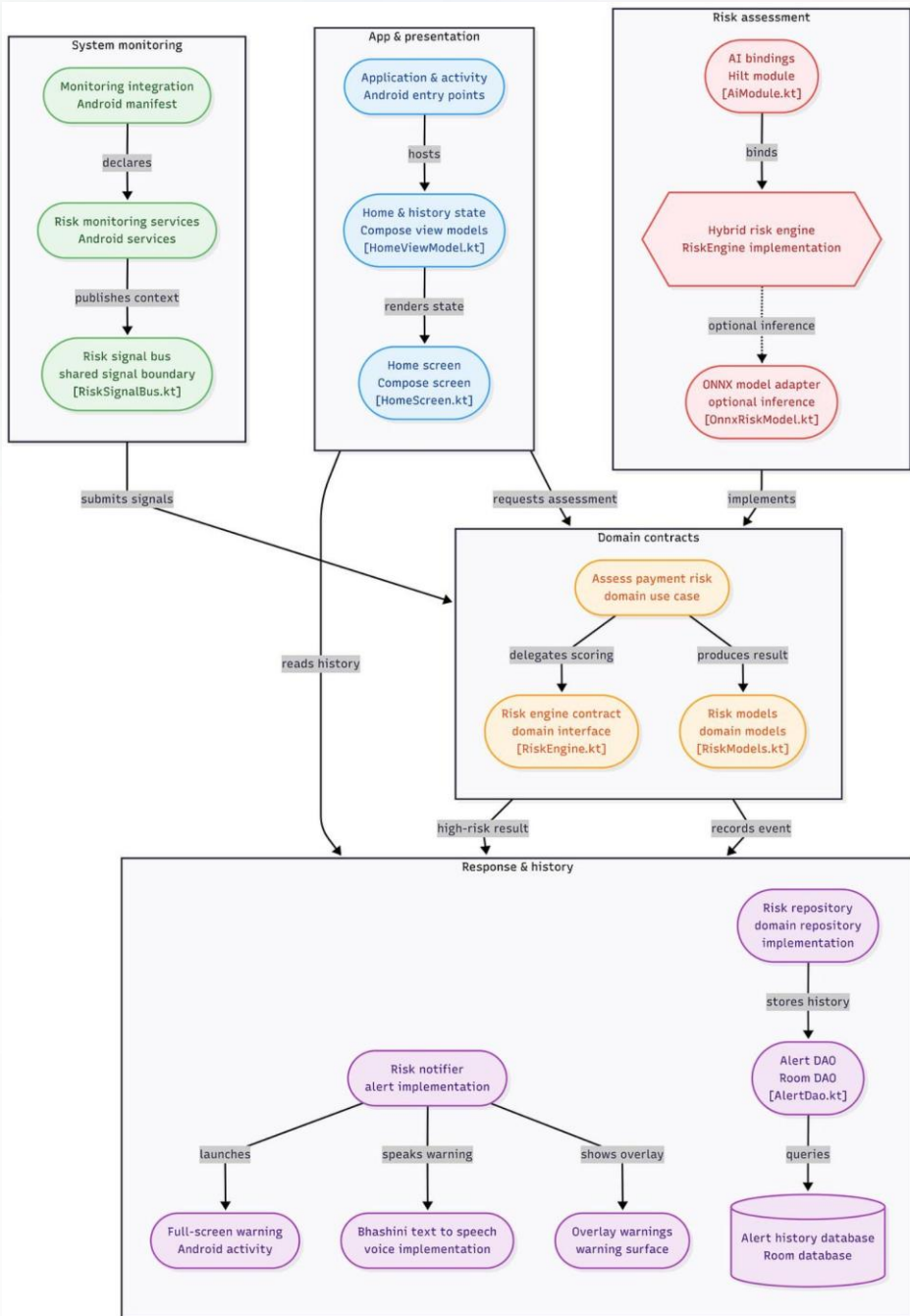
Bhashini Linguistic Bridge

- Bhashini AI API delivers real-time regional voice warnings.
Provides multilingual voice warnings to help users identify fraud before entering their UPI PIN.

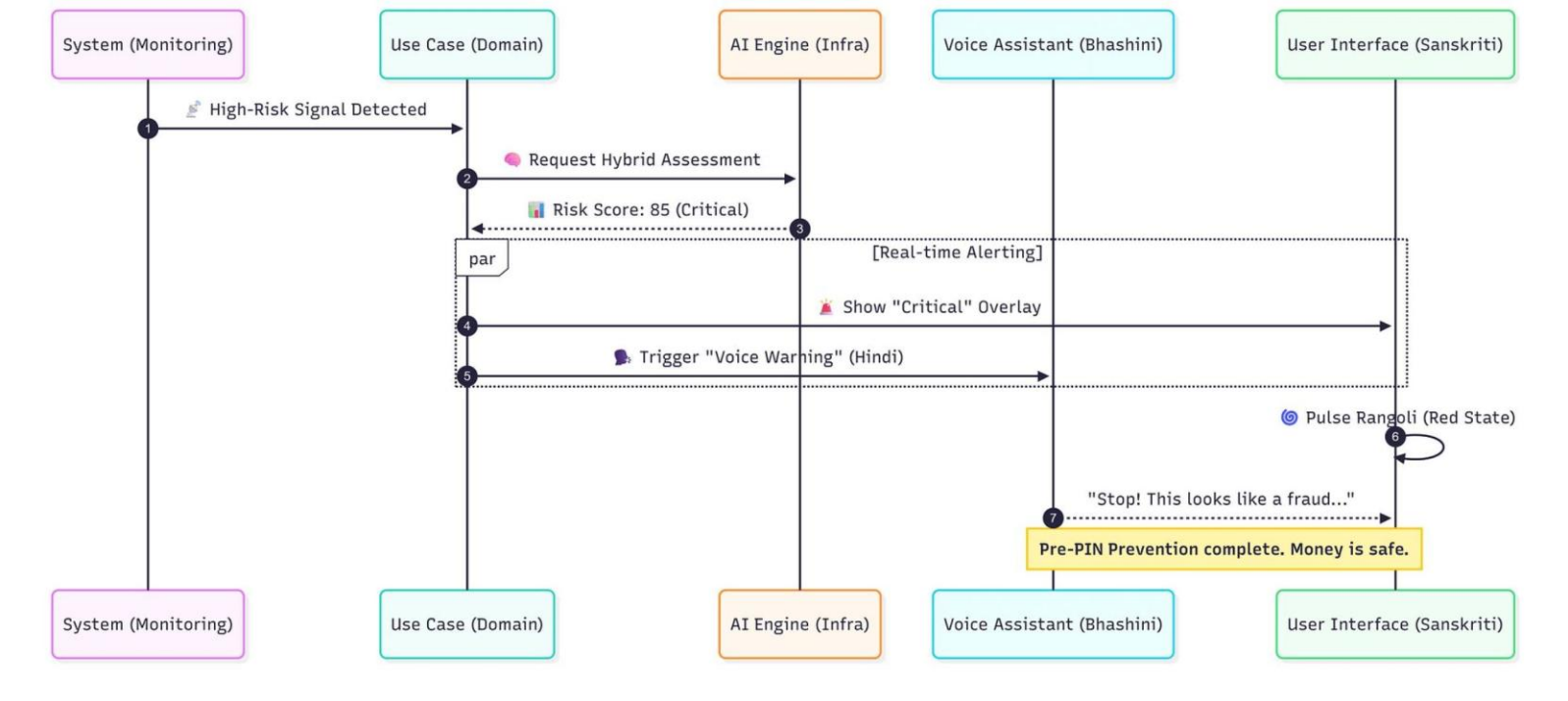
Privacy-Preserving Protocol

- All data processing happens locally on the device with no cloud dependency.
Processes all risk analysis locally on the device without cloud dependency.

Powered by Hilt • Room • Kotlin Coroutines • ONNX Runtime • Bhashini APIs



Real-Time Sequence Flow



System Monitoring

Continuously detects suspicious calls, SMS, URLs, and app activity.

AI Risk Engine

Hybrid AI evaluates risk and generates a real-time fraud score.

Real-Time Alerting

Critical alerts are displayed instantly before transaction authorization.

Bhashini Voice Assistant

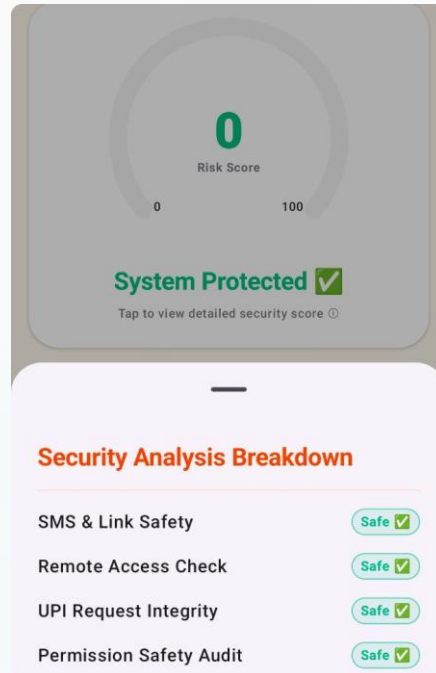
Bhashini provides multilingual voice warnings for better accessibility.

User Interface

User receives clear visual guidance to prevent fraudulent transactions.

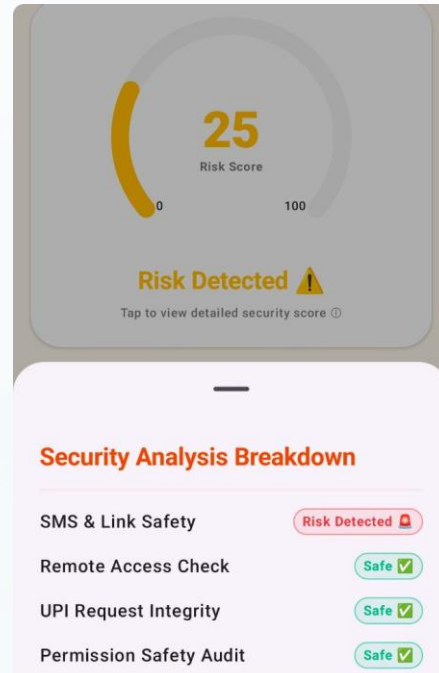
Real-time Risk Dashboard

Risk score is calculated in real time using Hybrid AI.



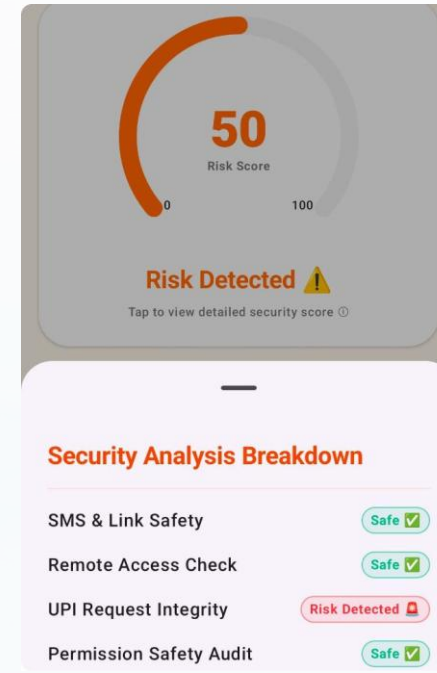
Safe (0/100)

No suspicious activity detected.



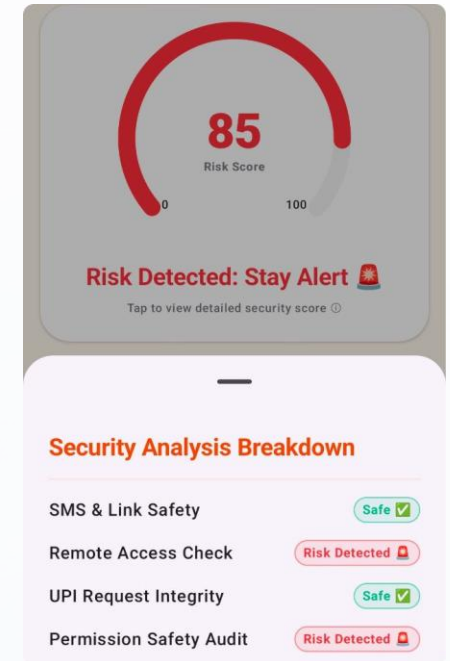
Caution (25/100)

Suspicious SMS or URL detected.



Moderate (50/100)

Potential UPI fraud indicators found.



Critical (85/100)

High-risk fraud detected.

App Onboarding & Guided Setup



1. Real-time Monitoring

GramKavach runs in the background and monitors for suspicious activities like unknown callers or remote access apps.

Real-time Monitoring

Runs in the background to monitor suspicious calls, SMS, URLs, and app activity.



2. Smart Risk Analysis

When a potential fraud is detected, you will see a risk score and a detailed breakdown of why the alert was generated.

Smart Risk Analysis

Analyzes suspicious activity and generates a real-time AI risk score.



3. Local Voice Alerts

Receive instant voice warnings in your own language to stay alert during high-pressure situations.

Local Voice Alerts

Provides multilingual voice warnings using Bhashini in the user's preferred language.



4. Emergency Reporting

Immediately report fraud attempts to the Cyber Cell by calling 1930 directly from the app.

Emergency Reporting

Quickly report suspected fraud by calling Cyber Helpline **1930** directly from the app.

"One-Time Setup. Continuous Protection"

THANK YOU

GramKavach CyberShield - Offline AI-Powered Protection for Digital Payments



 Live Demo / APK

Scan to experience the application

Questions & Discussion

 Offline AI

 Privacy-First

 Multilingual Support

 Real-Time Fraud Detection



 GitHub Repository

Scan to explore the source code

 *"Think Before You Tap. AI Already Did."*

Khushbu Prajapati • Government Engineering College, Patan • Maverick AI Effect Challenge 2026